Interior Permanent-Magnet Synchronous Motors for Adjustable-Speed Drives

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Abstract—**Interior permanent-magnet (IPM) synchronous motors possess special features for adjustable-speed operation which distinguish them from other classes of ac machines. They are robust high power-density machines capable of operating at high motor and inverter efficiencies over wide speed ranges, including considerable ranges of constant-power operation. The magnet cost is minimized by the low magnet weight requirements of the IPM design. The impact of the buried-magnet configuration on the motor's electromagnetic characteristics is discussed. The rotor magnetic circuit saliency preferentially increases the quadrature-axis inductance and introduces a reluctance torque term into the IPM motor's torque equation. The electrical excitation requirements for the IPM synchronous motor are also discussed. The control of the sinusoidal phase currents in magnitude and phase angle with respect to the rotor orientation provides a means for achieving smooth responsive torque control. A basic feedforward algorithm for executing this type of current vector torque control is discussed, including the implications of current regulator saturation at high speeds. The key results are illustrated using a combination of simulation and prototype IPM drive measurements.**

I. Introduction

A. Background

PERMANENT-magnet (PM) synchronous motors are attracting growing international attention for a wide variety of industrial applications, ranging from general-purpose line-start pump/fan drives [1] to high-performance machine tool servos [2]. The attractive power-density and efficiency characteristics exhibited by these motors as a class are major factors responsible for generating this interest. The recent announcements of more powerful and cost-effective permanent magnet materials are serving to accelerate these motor development efforts [3].

The large majority of commercially available PM synchronous motors are constructed with the permanent magnets mounted on the periphery of the steel rotor core, exposing their surfaces magnetically, and sometimes physically, to the air gap. These motors, referred to here as surface PM synchronous motors, are also known as brushless dc motors, inside-out motors, electronically commutated motors, as well as by a wide variety of manufacturer-specific trade names. This range of terminology obscures the fact that, in most cases, they are variations of the same class of machines.

Several interesting characteristics arise when the permanent magnets are mounted inside the steel rotor core. A sample geometry for this type of machine, known as the interior permanent magnet (IPM) synchronous motor, is shown in Fig. 1. Although this may at first seem to be a relatively modest variation of the surface PM geometry, the process of covering each magnet with a steel pole piece in the IPM geometry produces several significant effects on the motor's operating characteristics. For example, burying the magnets inside the rotor provides the basis for a mechanically robust rotor construction capable of high speeds since the magnets are physically contained and protected. In electromagnetic terms the introduction of steel pole pieces fundamentally alters the machine magnetic circuits, changing the motor's torque production characteristics. The nature of these changes and their beneficial consequences will be discussed at length in the body of this paper.

The basic IPM rotor configuration has been known for many years. The introduction of Alnico magnets nearly 50 years ago created a considerable interest in PM alternator development using interior PM motor geometries [4], [5]. Soft iron pole shoes in these alternators provided a means of concentrating the flux of the thick Alnico magnets. Improvements in PM materials in following years turned attention to integral-horsepower applications for PM synchronous motors. A combination of an induction motor squirrel cage and the interior PM geometry provided possibilities for efficient steady-state operation as well as robust line starting [6]. Work in this area accelerated during the past decade, following dramatic increases in the cost of energy [7].

Reports of variable-speed applications of interior PM synchronous motors also began to appear during the past decade. Most of this published work has originated in Europe, with Lajoie-Mazenc and his colleagues in France among the most active investigators [8], [9]. The IPM synchronous motor has also been explored in Europe for electric vehicle traction applications [10].

B. Scope of the Present Work

The purpose of this paper is to investigate the potential for achieving high-performance adjustable-speed operation by

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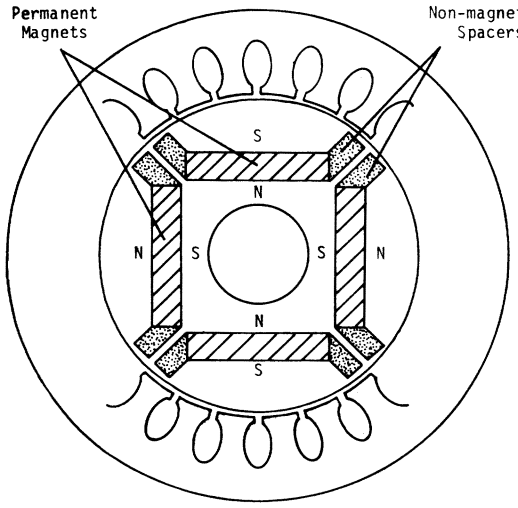


Fig. 1. Typical IPM synchronous motor lamination configuration.

combining an IPM synchronous motor with a transistorized inverter. Rather than describe a particular drive system, the objective of this paper is to identify and discuss more broadly the distinguishing features of the IPM synchronous motor for adjustable-speed operation. In the process the paper will draw on the collective experience of the authors with various motor designs and prototype drive systems tested to date.

Despite a desire to be as general as possible, the scope of the paper will be limited in at least two ways. First, the discussion will address IPM synchronous motors with radially oriented magnets based on the sample configuration in Fig. 1. Alternative buried-magnet motor designs, in which the magnets are mounted in the interpolar regions with circumferential magnetization [11], [12], share many generic characteristics but will not be specifically addressed in this paper. Second, the discussion will be limited to IPM synchronous motor drive systems supplied from voltage sources with regulation of the instantaneous motor phase currents, appropriate for high-performance applications. The implications of IPM synchronous motor operation with a classic current source inverter (i.e., ASCI-type) will be discussed only indirectly.

A sketch of a typical IPM synchronous motor drive power stage is provided in Fig. 2, consisting of a six-switch full bridge inverter which develops adjustable-frequency three-phase excitation from a dc voltage source (e.g., a line rectifier output or battery bank). The switches are illustrated as bipolar transistors, but any other bipolar- or MOS-based power switch device, which can be turned off as well as on from low-level gating commands, can also fill this role. Each switch is combined with a parallel freewheeling rectifier to provide circulation paths for the motor reactive phase currents. As shown in Fig. 2, it is assumed that the drive control electronics is provided with sensor feedback information from the three stator phase currents and the rotor position.

II. Motor Electromagnetic Characteristics

A. IPM Rotor Magnetic Circuit Saliency

In order to understand the operating characteristics of an IPM synchronous motor drive, it is necessary first to appreciate the distinguishing electromagnetic properties of the interior PM motor itself. In particular, it is important to recognize that burying the magnets inside the rotor introduces saliency into the rotor magnetic circuit which is not present in other types of PM machines.

By using the sample four-pole rotor geometry shown in Fig. 1, the magnetic flux induced by the magnets defines a direct or d axis radially through the centerline of the magnets; see Fig. 3(a). In the process an orthogonal quadrature or q axis is defined through the interpolar region separated from the d axis by 45 mechanical degrees (i.e., 90 electrical degrees for a four-pole design) as shown in Fig. 3(b). As sketched in Fig. 3(a) and (b), the magnetic flux passing through the d-axis magnetic circuit must cross two magnet thicknesses in addition to two air-gap crossings required in both the d and q axes. Since the incremental permeability of ceramic and rare-earth magnet materials is nearly that of free space, the magnet thicknesses appear as large series air gaps in the d-axis magnetic flux paths.

Since the q-axis magnetic flux in Fig. 3(b) can pass through the steel pole pieces without crossing the magnet air gaps, the stator phase inductance is noticeably higher with q-axis rotor orientation. The elevated permeance of the rotor q-axis magnetic circuit can be employed to enhance the adjustable-speed operating characteristics of IPM synchronous motors. For example, the additional inductance can be useful for depressing the required inverter switching frequency with the IPM synchronous motor compared to other types of ac machines, as demonstrated in Fig. 4. The relative magnitudes of the d- and q-axis inductance values depend on the details of the rotor geometry, and measured inductance ratios of three or higher have been reported in the literature [13].

The torque production in the IPM motor is altered as a result of the rotor saliency, providing design flexibility which can be exercised to shape the motor output characteristics beneficially. Note that the q-axis inductance of the IPM synchronous motor (L_q) typically exceeds the d-axis inductance (L_d), a feature which distinguishes the IPM motor from conventional wound-rotor salient-pole synchronous motors for which $L_d > L_q$. This reversal in the relative inductance values for the two axes has a direct effect on the torque production and excitation requirements for the IPM motor which will be discussed in the following sections.

B. Motor Equivalent Circuit and Torque Production

The magnetic saliency of the IPM synchronous motor rotor dictates that the electrical equivalent circuit be developed in the rotor reference frame. Standard assumptions regarding the sinusoidal stator winding distribution and the absence of iron saturation are made in order to carry out this development. By adopting the same orthogonal d and q axes defined in the preceding section, Park's transformation yields the classic two-axis equivalent circuit for a salient-pole synchronous motor [14] shown in Fig. 5. This is the same basic coupled-circuit pair used to model conventional wound-rotor salient-pole synchronous motors.

Although the derivation of this model is not included here, the significance of some of the important equivalent circuit elements deserves discussion. The rotor field excitation

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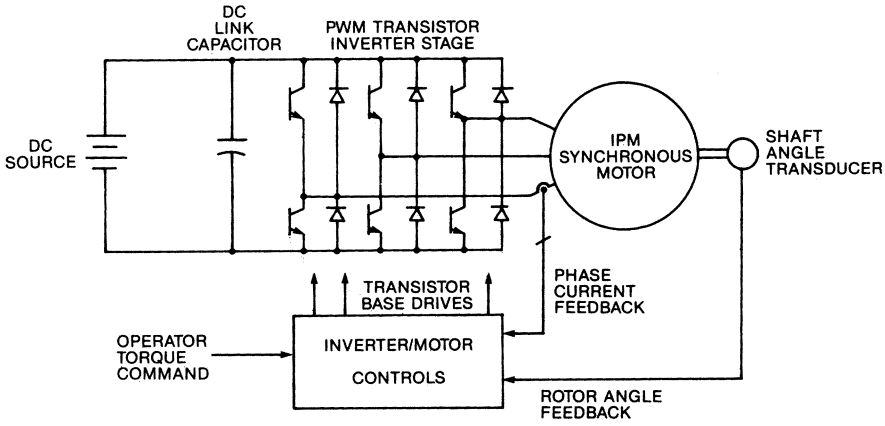


Fig. 2. Simplified schematic of IPM synchronous motor drive.

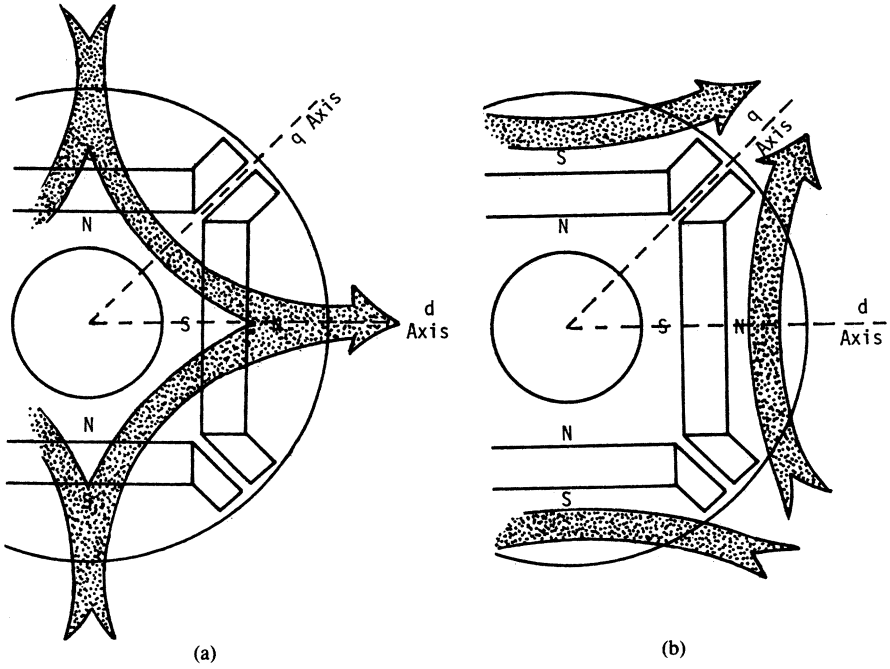


Fig. 3. Principal IPM magnetic flux paths. (a) d axis. (b) q axis.

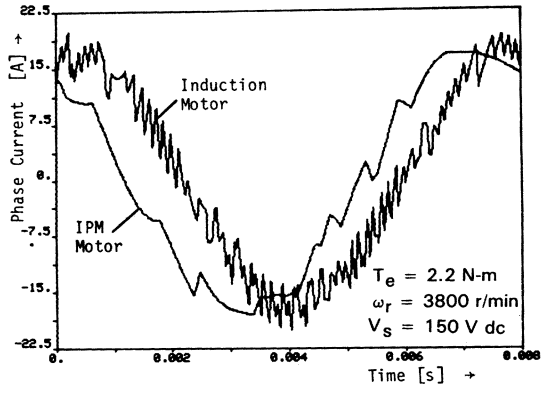
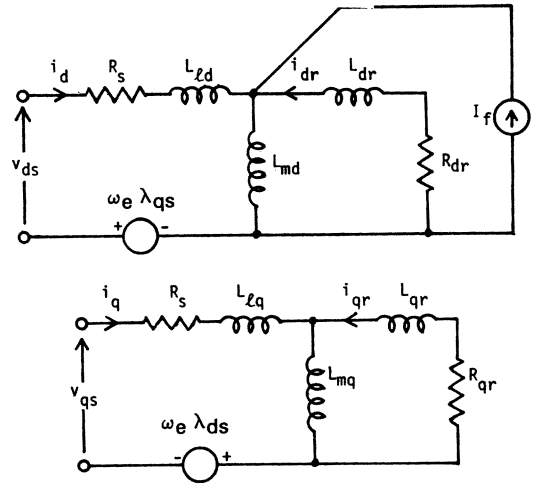


Fig. 4. Simulation results comparing IPM and induction motor phase current for equally rated 3-hp motors under identical load and supply test conditions with hysteresis-band current regulation.



$$\lambda_{ds} = (L_{\ell d} + L_{md})\, i_d + L_{md}\, i_{dr} + L_{md}\, I_f$$

$$\lambda_{qs} = (L_{\ell q} + L_{mq})\, i_q + L_{mq}\, i_{qr}$$

Fig. 5. IPM synchronous motor equivalent circuit in rotor reference frame.

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produced by the permanent magnets is modeled by an equivalent constant current source I_f, providing magnetizing flux $\Psi_{mag} = L_{md}I_f$ in the d axis. The higher permeance of the q-axis magnetic circuit is reflected in the distinct inductance elements in the two axis circuits such that $L_q(=L_{lq} + L_{mq})$ is larger than $L_d(=L_{ld} + L_{md})$. For completeness, the damper winding elements L_{dr}, R_{dr} and L_{qr}, R_{qr} are included in each of the axis circuits. These elements can be used to model discrete damper circuits purposely included in the rotor design [15] as well as distributed rotor eddy-current effects when deemed appropriate.

For steady-state operation when the damper transients have decayed to negligible levels, the average torque T_e developed by the IPM synchronous motor can be expressed in terms of the Fig. 5 equivalent circuit d–q currents as

$$T_e = 1.5p[I_{qs}\Psi_{mag} + (L_d - L_q)I_{qs}I_{ds}] \quad (1)$$

where

Ψ_{mag} permanent magnet flux linkage ($=L_{md}I_f$),
L_d, L_q total d axis ($=L_{md} + L_{ld}$) and q-axis ($=L_{mq} + L_{lq}$) stator inductances,
p number of pole pairs,
I_{qs}, I_{ds} steady-state q-axis and d-axis stator currents.

Each of the two terms in this equation reflects an important aspect of the torque production in an IPM synchronous motor. First, the magnet flux oriented along the rotor d axis interacts with the q-axis stator current to produce a field-alignment torque proportional to the ($\Psi_{mag}\, I_{qs}$) product. This is the same process by which torque is produced in a conventional surface PM synchronous motor. In addition, the current-induced magnetic fluxes along the two axes L_dI_{ds} and L_qI_{qs} interact with the orthogonal current components to contribute a second torque term. The rotor saliency is clearly responsible for the presence of this reluctance torque term, which is proportional to the axis inductance difference ($L_d - L_q$). Thus the torque equation suggests that, for purposes of conceptualization, the IPM motor can be interpreted as a hybrid combination of the conventional synchronous-reluctance and surface PM machines.

The IPM drive system performance characteristics can be influenced by adjusting the IPM rotor design parameters to control the relative contributions of the field-alignment and reluctance torque terms. For example, overexcitation conditions in a PM synchronous motor drive pose potential dangers to the drive electronics when the magnet-generated motor back EMF significantly exceeds the source voltage at high speeds. The rotor saliency can be employed to reduce the PM excitation flux requirements in the IPM motor in order to achieve extended-speed operating ranges while proportionally reducing the overexcitation amplitude and its attendant risks. From an economic standpoint, rotor saliency provides opportunities for reducing the volume of magnet material in the IPM motor which would otherwise be required to achieve a desired motor power rating.

C. Effect of Iron Saturation

The nonlinear performance effects introduced by iron saturation in any ac machine are further complicated in the IPM synchronous motor by the salient rotor magnetic circuits. When the MMF contributions of the rotor magnets and the d- and q-axis stator current components are summed, the resulting unsaturated air-gap flux distribution shown in Fig. 6 has a distinctly nonsinusoidal waveshape [16]. The elevated magnetic permeance of the rotor q axis provides conditions for high magnetic flux densities at the edges of the iron pole pieces. As a result, the stator teeth opposite the leading edges of these poles are particularly vulnerable to iron saturation as the current excitation level is raised.

The saturation of these segments of the stator teeth has the effect of reducing the fundamental spatial component of the air-gap flux density for a given stator current and shifting it toward the center of the pole. From the terminals of the motor this air-gap flux reduction appears as a reduction in the stator inductances, particularly in the higher permeance q-axis circuit. The inherent nonlinear nature of these saturation effects, combined with the salient rotor structure, creates cross-coupling effects in the two flux axes, which pose difficult modeling problems beyond the scope of this paper [16], [17]. However, it is clear that iron saturation typically serves to linearize the torque versus stator current relationship at higher currents, compared to the ideal case without saturation as shown in Fig. 7.

D. Motor Losses and Efficiency

An attractive performance characteristic which the IPM synchronous motor shares with other types of permanent magnet ac motors is its high electrical efficiency. The rotor losses in the IPM motor are significantly lower than in a comparable induction motor, since no current-carrying windings exist on the rotor to accumulate resistive I^2R losses. The reductions in the rotor losses are particularly valuable since losses are almost always more difficult to thermally extract from a spinning rotor than from the surrounding stator.

Tests with prototype IPM synchronous motors have confirmed their very attractive power density and loss characteristics compared to other types of ac machines. For example, a 3-hp prototype IPM synchronous motor tested at its rated speed of 4800 r/min has demonstrated a full-load efficiency in excess of 94 percent. Since ferrite magnets are used in this particular machine, confidence exists that such efficiency numbers will be pushed still higher in future motors designed with new generations of high-energy-product neodymium–iron magnets [3].

III. IPM Adjustable-Frequency Excitation Issues

A. Basis of Instantaneous Torque Control

A prerequisite for high-performance velocity or position control in all adjustable-speed ac drives is responsive control of the instantaneous torque. In particular, it is vital to minimize the sources of pulsating torque in order to prevent undesired pulsations in the rotor speed. This requirement has a significant effect on the techniques for achieving instantaneous torque control in an IPM synchronous motor.

The torque production in any ac motor can be interpreted as resulting from the interaction of the air-gap magnetic flux density distribution and the stator current MMF distribution

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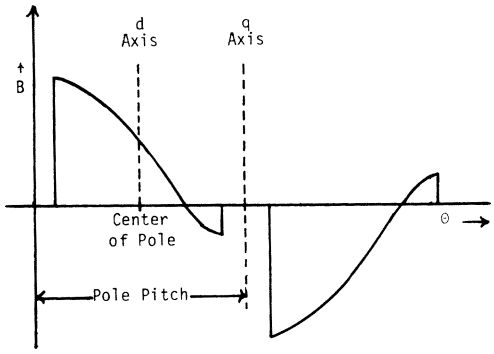


Fig. 6. Nominal IPM air-gap magnetic flux density distribution.

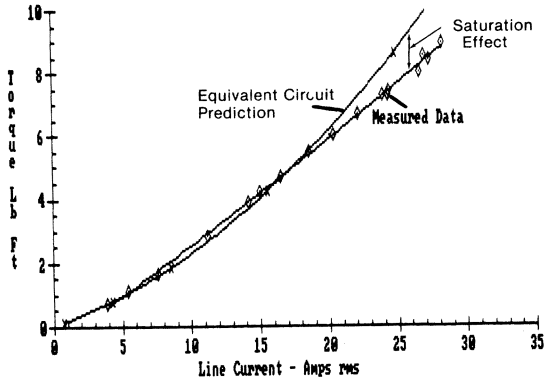


Fig. 7. Comparison of linear equivalent circuit model steady-state torque prediction with measured test results for 3-hp prototype IPM drive.

along the stator air-gap surface. As shown in Fig. 6, the air-gap flux density distribution in the IPM motor is distinctly nonsinusoidal. Under these conditions the most convenient way of producing a smooth constant torque is to generate a synchronously rotating stator current MMF wave which is fixed in space relative to the rotor surface. This requirement for a uniform traveling MMF wave strongly suggests balanced sinusoidal excitation of the three-phase stator windings which, by assumption, are sinusoidally distributed.

Conversely, square wave excitation will not meet the conditions for smooth torque generation in the IPM motor, since the square waves will produce an MMF wave which discretely shifts along the air gap only at the switching instants. This unacceptability of square wave excitation distinguishes the IPM synchronous motor from its surface-magnet counterpart which can be designed for sinusoidal or square-wave excitation [18].

The control of the instantaneous phase currents provides a direct means of controlling the instantaneous torque developed by the motor. This becomes particularly apparent when the motor is designed to minimize all rotor damper effects (see Fig. 2), because without dampers the torque equation (1) applies for instantaneous values of the torque and current as well as for the average values. That is, the removal of the damper effects causes the torque to respond immediately to changes in the stator current components i_d and i_q without dynamic terms associated with the damper transients. Since the absence of these dynamics permits valuable simplifications of the torque control algorithm described in the following sections, it will be assumed that rotor damper effects are negligible for the remainder of this paper.

Several pulsewidth modulation (PWM) techniques have been developed to provide control of the instantaneous phase currents for any polyphase ac machine [19], [20]. Although these algorithms will not be described here, it must be noted that sinusoidal control of the three-phase currents typically requires current sensors in series with the individual phase windings. In addition, the sinusoidal excitation of the IPM synchronous motor requires rotor angle feedback with sufficient resolution to synchronize the sinusoidal references properly with the rotor position. These requirements are generally more demanding than for comparable six-step square-wave current excitation configurations for which the rotor angle information is necessary only in 60° increments.

B. Stator Current Vector Control

The relationships between the stator phase current amplitudes and the instantaneous torque can be conveniently described with the aid of vector notation. Fig. 8(a) shows the three stator phase axes defined at 120° intervals, with two motor poles assumed for simplicity. If each scalar phase current is depicted as a magnitude-scaled vector along its appropriate axis (or negative axis for negative current values), the three component phase current vectors can be vectorially summed to form the resultant stator current vector $\vec{i}_s$ shown in Fig. 8(a). Note that all of the currents are instantaneous values. For steady-state balanced excitation, vector $\vec{i}_s$ will have a constant amplitude and rotate at the excitation angular frequency ω_e.

Fig. 8(b) shows how this stator current vector $\vec{i}_s$ can be usefully related to the rotor. The instantaneous position of the rotor d axis defined by the rotor magnet flux (see Fig. 3) is at an angle θ_r with respect to the phase A stator axis. At every instant the stator current vector can be decomposed into its two orthogonal components i_d and i_q along the rotor d and q axes as shown in Fig. 8(b). For a fixed stator current magnitude, i_d and i_q become constant values when the angular velocity of the current vector is forced to match that of the rotor. This synchronization of excitation and rotor speeds satisfies a necessary condition for smooth instantaneous torque production in a synchronous machine.

Fig. 9 shifts the viewpoint from the stator reference frame depicted in Fig. 8 to the rotor reference frame fixed to the rotor d and q axes. Assuming that the damper effects are made negligible by design, the relationship between the instantaneous stator current components (i_d and i_q) and the torque T_e is expressed by (1). Within the limits of iron saturation, this equation defines a hyperbola of (i_d, i_q) couples in the rotor reference frame for every value of torque. Fig. 9 shows the resulting curve for one particular value of positive torque along with three of the infinite number of stator current vectors which would deliver this same torque.

A closer examination of (1) reveals that useful insights can be gained from normalization as follows:

$$T_{en} = i_{qn}(1 - i_{dn}) \tag{2}$$

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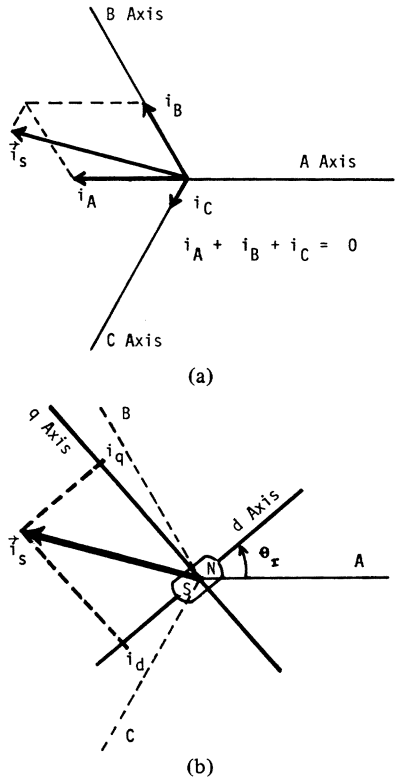


Fig. 8. Instantaneous current vector definition. (a) In terms of stator phase currents. (b) In rotor reference frame, including $i_d - i_q$ decomposition.

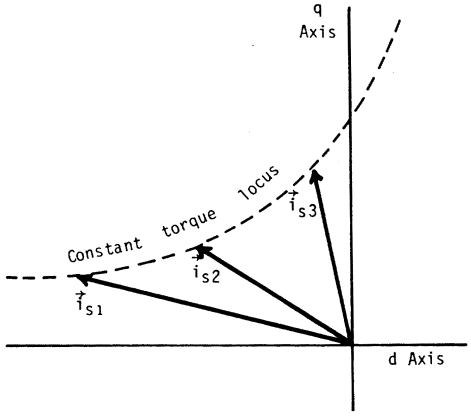


Fig. 9. Typical constant torque locus for IPM synchronous motor in rotor reference frame showing three sample stator current vectors delivering same electromagnetic torque.

where

$$T_{en} = \frac{T_e}{T_{eb}} \qquad i_{qn} = \frac{i_q}{i_b} \qquad i_{dn} = \frac{i_d}{i_b}$$

$$i_b = \frac{\Psi_{\text{mag}}}{(L_q - L_d)}$$

$$T_{eb} = 1.5p\Psi_{\text{mag}} i_b.$$

All of the motor parameters are eliminated from the resulting normalized torque–current relationship. Fig. 10 provides a family of curves in the normalized i_{qn}, i_{dn} plane for several different values of normalized torque. Normalization allows these curves to apply to any combination of IPM motor parameter values within the linearity limits imposed by iron saturation, etc. In addition to the symmetry, note that T_{en} is positive throughout the second quadrant (motoring torque for counterclockwise (CCW) rotation) and negative in the third quadrant (braking torque for CCW rotation).

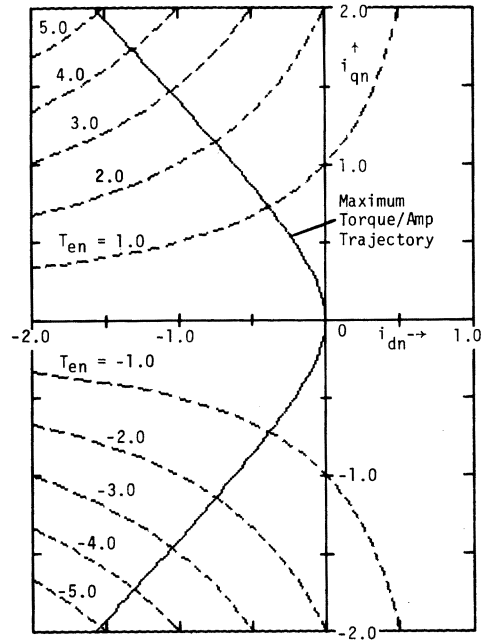


Fig. 10. Constant torque loci for IPM synchronous motor in terms of normalized phase current and torque variables. Current vector trajectory for maximum torque/ampere is also plotted.

Since a particular value of torque can be developed with an infinite set of distinct (i_{dn}, i_{qn}) combinations, a question naturally arises regarding the optimal choice of i_{dn} and i_{qn} as T_{en} varies. If motor efficiency is an important performance characteristic, one attractive optimization criteria is maximum torque per stator current ampere. Fig. 10 includes the i_{dn}, i_{qn} trajectory of maximum torque/ampere for positive and negative torque values. Note that each trajectory–torque curve intersection represents the point on that particular curve which is closest to the origin, corresponding to a minimum stator current. Fig. 11 provides plots of the i_{dn} and i_{qn} coordinates for the maximum torque/ampere trajectory as a function of the normalized torque. These trajectories are defined (using primed variables) by the following equations:

$$T'_{en} = \sqrt{i'_{dn}(i'_{dn} - 1)^3} \tag{3}$$

$$T'_{en} = \frac{i'_{qn}}{2}\left[1 + \sqrt{1 + 4(i'_{qn})^2}\right]. \tag{4}$$

Some interesting insights into the IPM synchronous motor torque production are found by examining the details of this maximum torque/ampere trajectory. First, note that the trajectory in Fig. 10 is tangent to the q axis at the origin and asymptotes to 45° trajectories in both the second and third quadrants. This clearly reflects the hybrid nature of torque production in the IPM motor, since the q axis represents the

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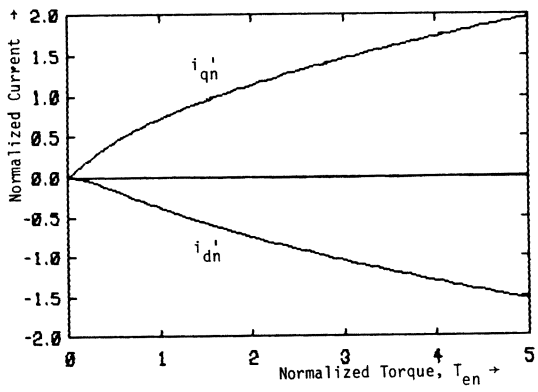


Fig. 11. Calculated normalized stator current components as function of normalized torque for maximum torque/ampere trajectory. Iron saturation effects neglected.

optimal trajectory for the field-alignment torque alone while the 45° asymptotes correspond to the reluctance torque term ($L_q > L_d$). As the torque is increased, the reluctance torque term, proportional to the square of the current, increasingly dominates the field-alignment torque term, which is only linearly proportional to current. This hybrid quality is also reflected in (4) where $T'_{en} \approx i'_{qn}$ for low current values and $T'_{en} \approx (i'_{qn})^2$ for high currents.

Although the preceding discussion is idealized since it strictly applies only for constant motor parameters, further study has indicated that all of the key observations hold in the presence of moderate iron saturation. As a result of the localized stator teeth saturation associated with the Fig. 6 flux distribution, the maximum torque/ampere trajectory tends to shift toward the q axis as the stator current is increased. In addition, the iron saturation tends to linearize the torque–current relationship at high currents as shown in Fig. 7.

C. Feedforward Torque Control Configuration

At this point all of the key concepts necessary to design a high-performance torque controller for the IPM synchronous motor have been introduced. Although a wide range of alternative designs might be proposed, a simple feedforward torque controller configuration will be discussed for illustration purposes. Besides simplicity, the feedforward controller shown in Fig. 12 has the advantage of requiring only phase current and rotor position feedback. However, the feedforward nature of the controller demands that the motor characteristics be directly reflected in the function blocks $f_1(T_e^*)$ and $f_2(T_e^*)$. (The asterisks denote the commanded values.)

As discussed in preceding sections, minimizing the rotor damper effects results in the elimination of the dynamic terms from the stator current–torque relationship. Thus the function blocks f_1 and f_2, which convert the incoming torque requests into the required stator current component commands i_d^* and i_q^*, can be simple time-independent function generators. Although an infinite number of candidates exist for f_1 and f_2, the curves in Fig. 11 provide attractive choices if high motor efficiency with maximum torque/ampere is important.

The vector rotator stage converts the i_d^* and i_q^* commands into equivalent phase current commands i_A^*, i_B^*, and i_C^*, requiring a coordinate transformation from the rotor reference frame to the stator frame. This operation requires information on the instantaneous rotor position θ_r to ensure proper synchronization at all times. By using rotor position sensor feedback to perform this synchronization, no danger exists of pole slippage between the excitation and rotor position regardless of loading conditions. The defining trigonometric relations are given by

$$i_A^* = i_d^* \cos(\theta_r) - i_q^* \sin(\theta_r) \tag{5}$$

$$i_B^* = i_d^* \cos(\theta_r - 120°) - i_q^* \sin(\theta_r - 120°) \tag{6}$$

$$i_C^* = i_d^* \cos(\theta_r + 120°) - i_q^* \sin(\theta_r + 120°). \tag{7}$$

These instantaneous phase current commands are then amplified and applied to the motor phase windings by means of the power converter stage, using phase current feedback to provide PWM closed-loop current regulation.

The dynamic response characteristics of the IPM synchronous motor drive with this type of feedforward torque control scheme are compatible with the requirements of many high-performance applications. The digital simulation results presented in Fig. 13 illustrate a typical IPM drive system response to a large-signal step in the torque request. The motor parameters for this simulation have been drawn from a prototype 5-hp 2200 r/min prototype IPM synchronous motor.

Fig. 13 indicates that the rise time for the instantaneous torque is less than 1 ms for these typical conditions. The residual high-frequency pulsations in the currents and torque are associated with the PWM switching which executes the current regulation. Note that the d-axis stator current i_d responds more rapidly than the q-axis current i_q, which is consistent with the lower d-axis inductance value. Although rotor dampers might be introduced to accelerate these responses, the adverse damper effects on the inverter switching frequency and losses demand special trade-off considerations which will not be discussed here. All of the important variable responses in Fig. 13 are well-behaved, as confirmed by laboratory tests.

D. Six-Step Saturated Regulator Operation

The finite dc bus voltage is responsible for imposing limits on the drive system torque–speed operating envelope at high speeds. The nature of this limit can be understood by noting that for any given values of the stator current components i_d and i_q (and thus torque), the stator voltage vector amplitude is nearly proportional to speed. When the resulting line-to-line terminal voltage approaches the fixed dc bus voltage as the speed is increased, the driving voltage necessary to force the stator currents to their commanded values decays to zero. Under these conditions the current regulators saturate, the pulses in the phase voltage waveforms drop out as the PWM current control is lost, and the system eventually reverts to six-step voltage excitation.

Fig. 14 presents some typical IPM motor phase voltage and current waveforms measured during the six-step voltage

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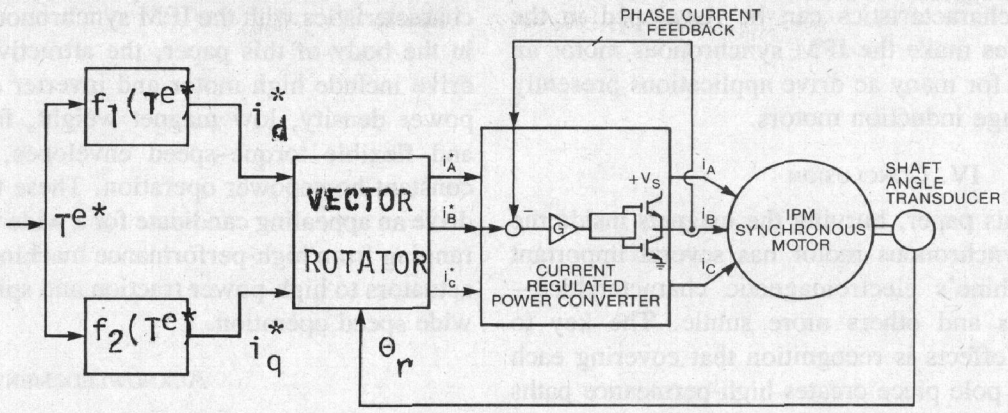


Fig. 12. Feedforward torque control block diagram for IPM synchronous motor drive.

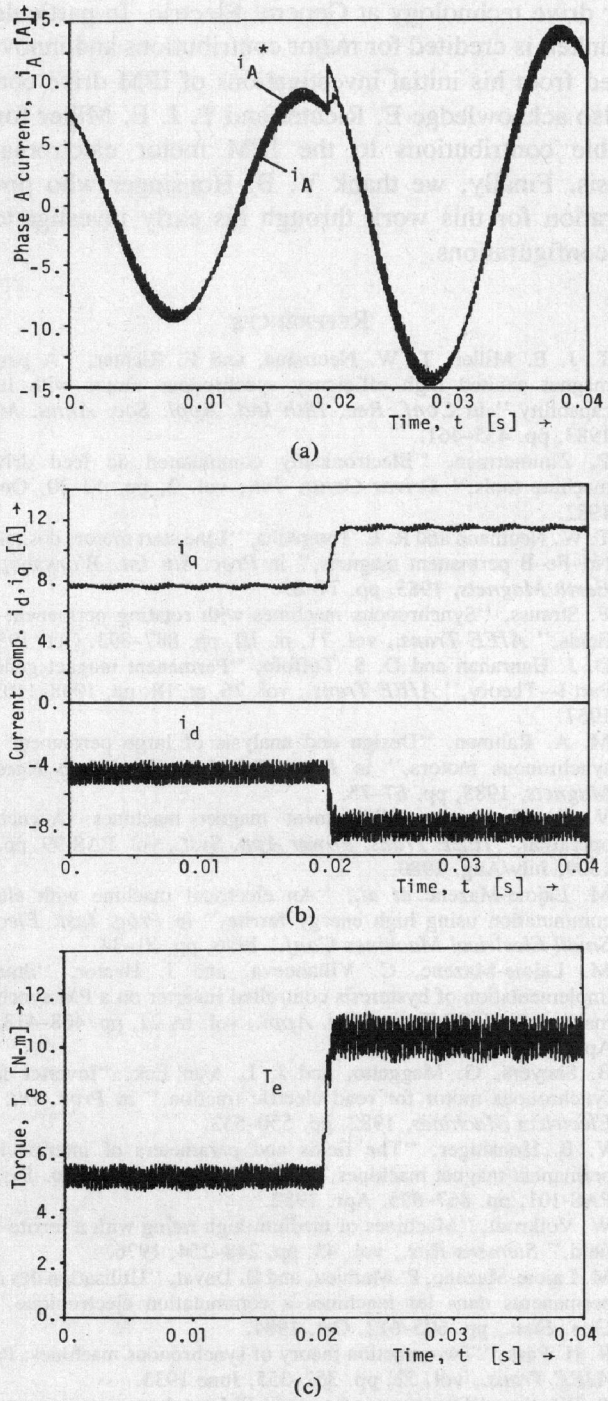


Fig. 13. Transient response simulation of IPM drive using feedforward torque controller to large-signal torque command step. Parameters from 5-hp prototype drive operated at 1000 r/min. (a) Phase A current. (b) i_q and i_d. (c) Torque.

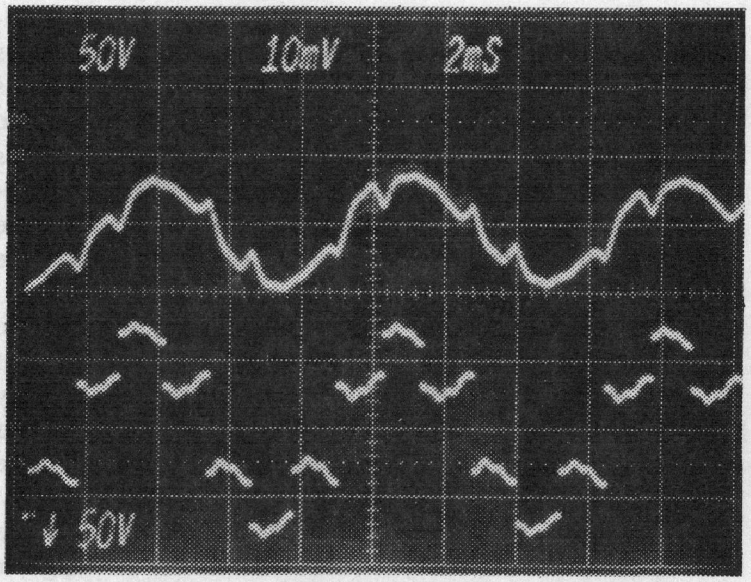


Fig. 14. Measured six-step excitation phase current and phase voltage waveforms for 3-hp prototype IPM drive at 4100 r/min. Upper: i_A – 10 A/div. Lower: v_{An} – 50 V/div. Horizontal: t – 2 ms/div.

excitation of a 3-hp prototype drive system. Although the phase currents are no longer regulated to follow the sinusoidal references, the elevated phase inductances of the IPM motor serve the useful purpose of filtering the six-step voltage harmonic components, thereby limiting the periodic current peaks. These current peaks are undesirable because of the their adverse effects on the peak current ratings of the inverter switches, inverter switching losses, and pulsating torque components.

The saturation of the current regulators with the onset of six-step voltage excitation requires the nature of the IPM drive torque control to change from current control to voltage control. This transition typically entails some degradation in the torque control characteristics, since only the voltage vector angle and not the amplitude can be adjusted during six-step excitation. The availability of rotor position feedback at all speeds makes it possible to control this voltage vector flexibly angle during six-step operation without any danger of pole slippage (pullout), just as during regulated-current operation.

Although this voltage control mode will not be discussed in detail in this paper, note that six-step voltage excitation can significantly expand the achievable torque–speed operating envelope of the IPM drive. Consider, as an example of constant-

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horsepower output characteristics can be developed in the process. Such features make the IPM synchronous motor an attractive alternative for many ac drive applications presently served by squirrel-cage induction motors.

IV. Conclusion

As described in this paper, burying the magnets inside the rotor of the IPM synchronous motor has several important effects on the machine's electromagnetic characteristics—some rather obvious and others more subtle. The key to understanding these effects is recognition that covering each magnet with an iron pole piece creates high-permeance paths for the magnetic flux across these poles and orthogonal to the magnet flux. The effects of this saliency show up directly in the IPM torque equation where, in addition to the field-alignment term common to the surface-magnet synchronous motor, a second reluctance torque term exists which is dependent on the magnetic permeance difference in the two orthogonal rotor axes. Furthermore, the IPM motor is distinguished from conventional wound-rotor salient synchronous machines by the fact that the IPM stator phase inductance with direct-axis (magnet) alignment L_d is less than the quadrature-axis inductance L_q.

The same six-switch full-bridge inverter used to excite the induction motor and surface PM synchronous motor can also be used to achieve high-performance adjustable-frequency operation with an IPM synchronous motor. Key insights and observations regarding the adjustable-frequency performance characteristics of the IPM motor include the following.

1) The basis of high-performance instantaneous torque control with the IPM motor is control of the angular orientation of the stator phase excitation with respect to the rotor position at all times. Rotor position transducer feedback is the standard means of providing this self-synchronization function, ensuring that excitation pole slippage (pullout) will never occur. This basic angle control of the IPM synchronous motor should be distinguished from the frequency control (i.e., slip frequency) incorporated into familiar induction motor control algorithms.

2) In particular, closed-loop regulation of the motor phase currents provides an attractive means of achieving responsive instantaneous torque control with the IPM synchronous motor. By exciting the motor with balanced sinusoidal current waveforms synchronized with the rotor, torque pulsations will be eliminated at all speeds in spite of nonlinearities in the spatial air-gap magnetic flux distribution.

3) A wide spectrum of alternative algorithms can be developed for executing torque control in the IPM motor by means of phase current regulation. One particularly straightforward candidate has been described which uses feedforward control to convert directly an incoming torque command into rotor-referenced stator current commands i_d^* and i_q^*, according to predefined functions. Simulation (Fig. 13) and measured prototype results have confirmed that responsive torque control is achievable with the IPM synchronous motor using only rotor position and phase current feedback.

The adoption of these excitation control principles makes it possible to achieve high-performance adjustable-speed drive characteristics with the IPM synchronous motor. As described in the body of this paper, the attractive features of the IPM drive include high motor and inverter efficiency, high motor power density, low magnet weight, fast dynamic response, and flexible torque–speed envelopes, including high-speed constant-horsepower operation. These features make the IPM drive an appealing candidate for a wide variety of applications, ranging from high-performance machine tool servos and robot actuators to high-power traction and spindle drives demanding wide speed operation.

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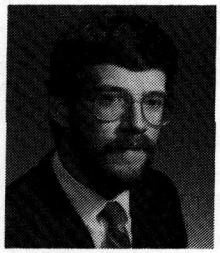
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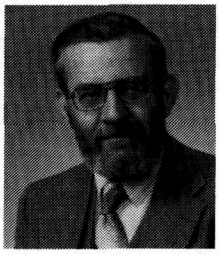
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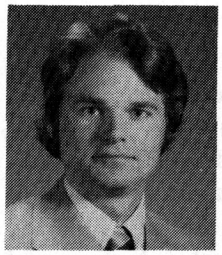
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